Jul–Aug 2026

BRIC Research Centre

FROM NETWORK TO DRUG TARGET

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Recap: Previous session

- MCODE gave us a ranked list of communities, but it told us nothing about what the proteins involved do.
- CytoHubba gave us ranked lists of proteins based on different metrics, but it also told us nothing about the biological processes these proteins are involved in.
- Both outputs, as they stand, are just lists of gene names. Today's session is about attaching meaning to those lists, a functional label, a pathway position, and a druggability check.

What is enrichment?

- Input: List of entities, proteins, genes, metabolites,...
- Enrichment analysis asks a single question of that list: are these genes involved in the same biological process/function more often than random chance would predict?
- Output: a list of statistically over-represented terms. Enrichment is not just a list of terms simply "present" in your gene list, rather, enrichment specifically means more present than chance alone would explain.

Gene Ontology (GO)

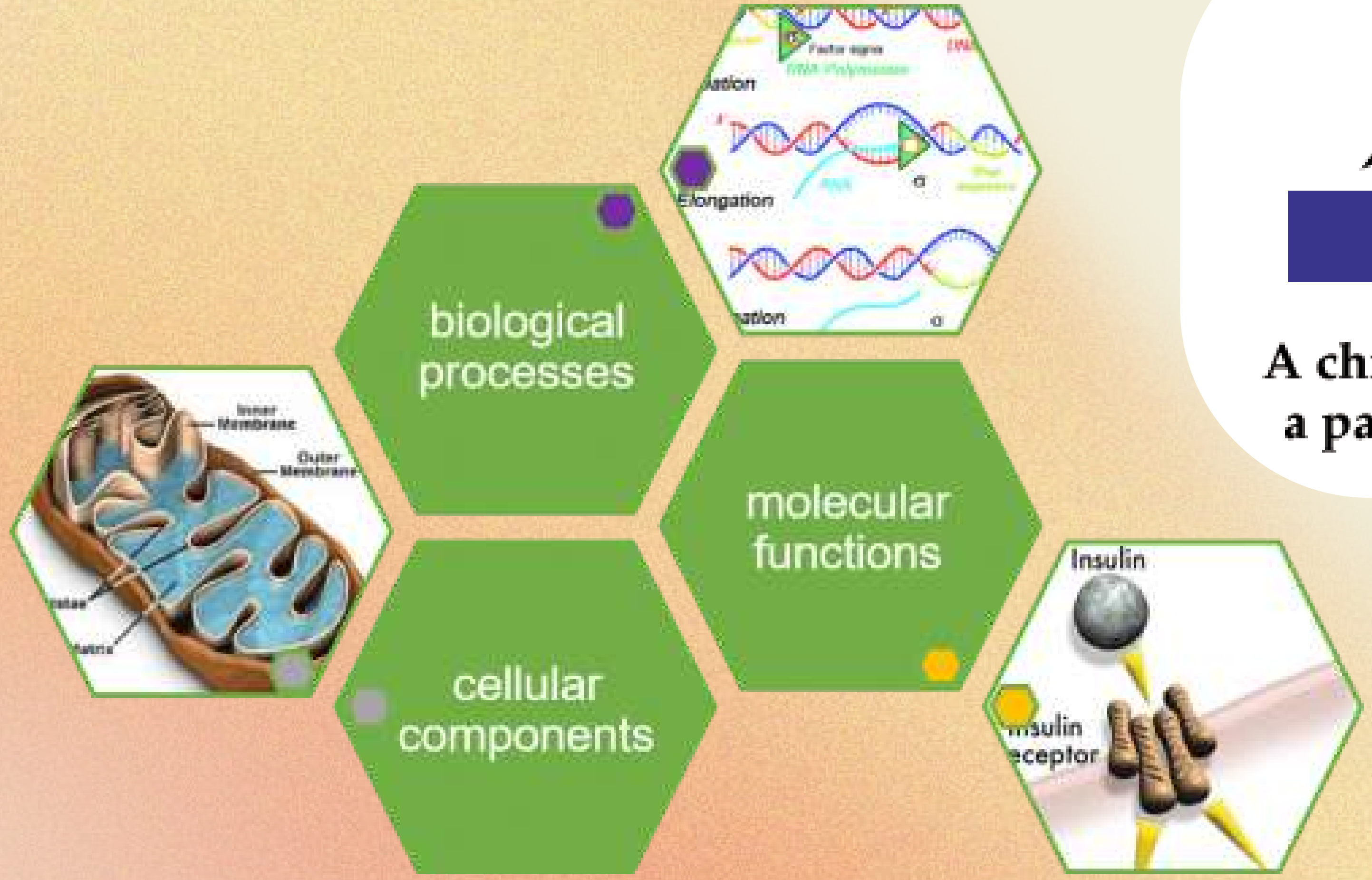
- GO is a structured, hierarchical controlled vocabulary describing gene product properties, maintained by the Gene Ontology Consortium.

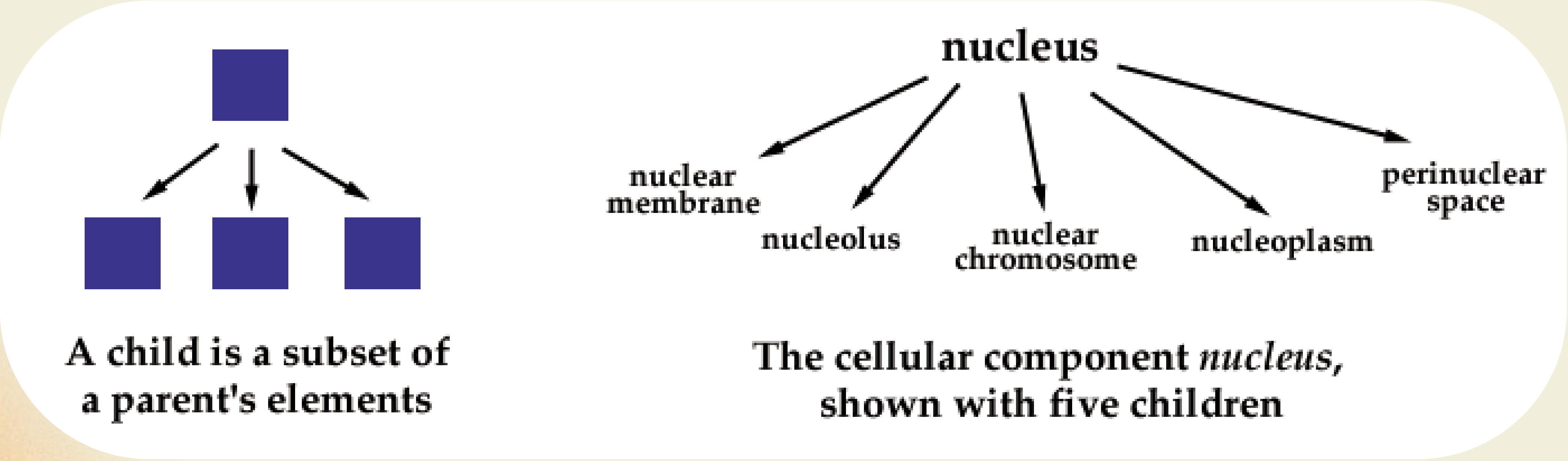
Three categories:

- Biological Process (BP): The larger process the gene product participates in (e.g., "inflammatory response").
- Molecular Function (MF): The biochemical activity (e.g., "kinase activity"). Describes what the protein does, independent of the larger context it does it in.
- Cellular Component (CC): Physical location within the cell (e.g., "plasma membrane").

Gene Ontology (GO)

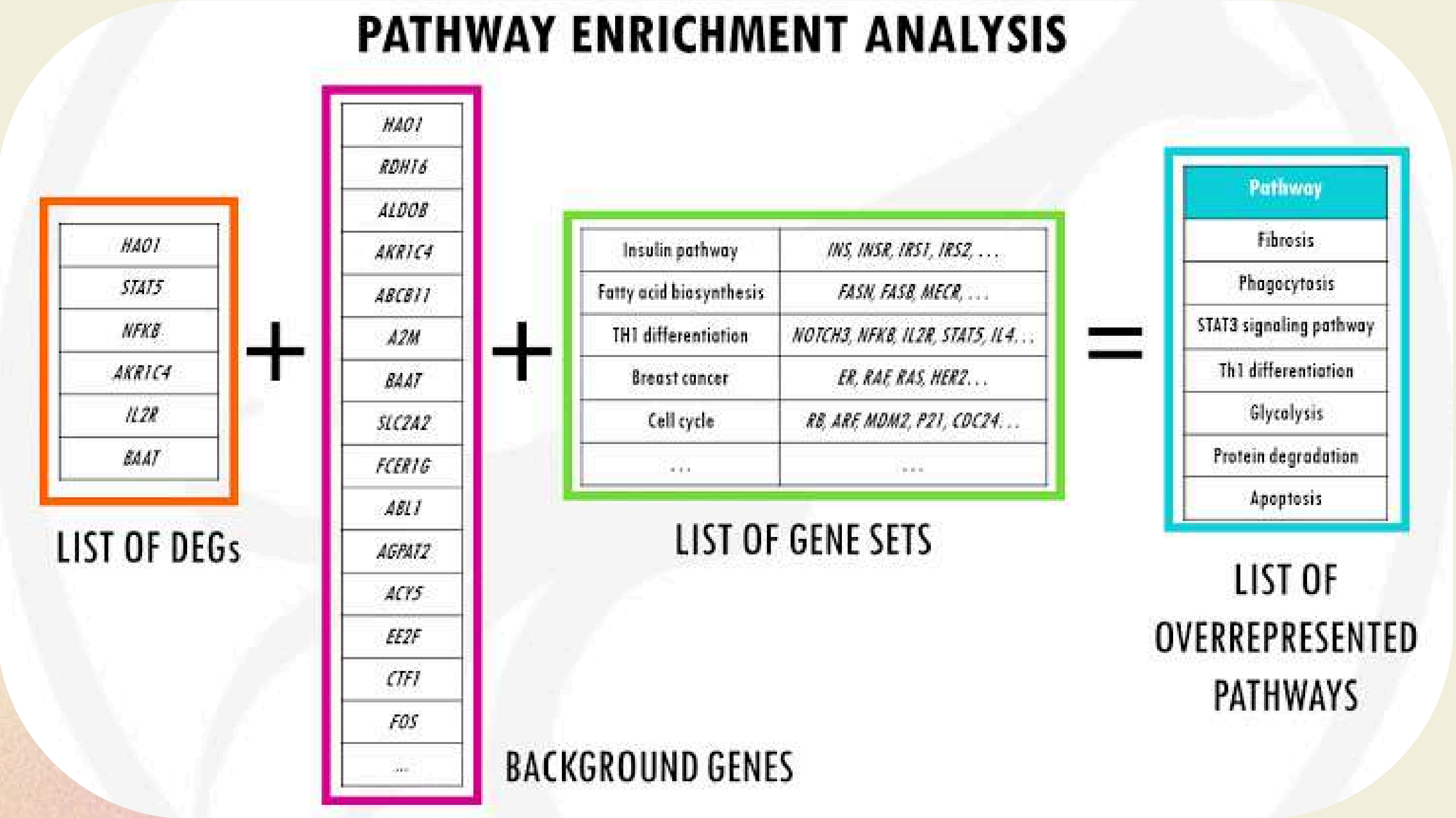
- Terms form a hierarchy from broad parent terms down to specific child terms, i.e. a gene annotated to a child term is implicitly annotated to all its parents.





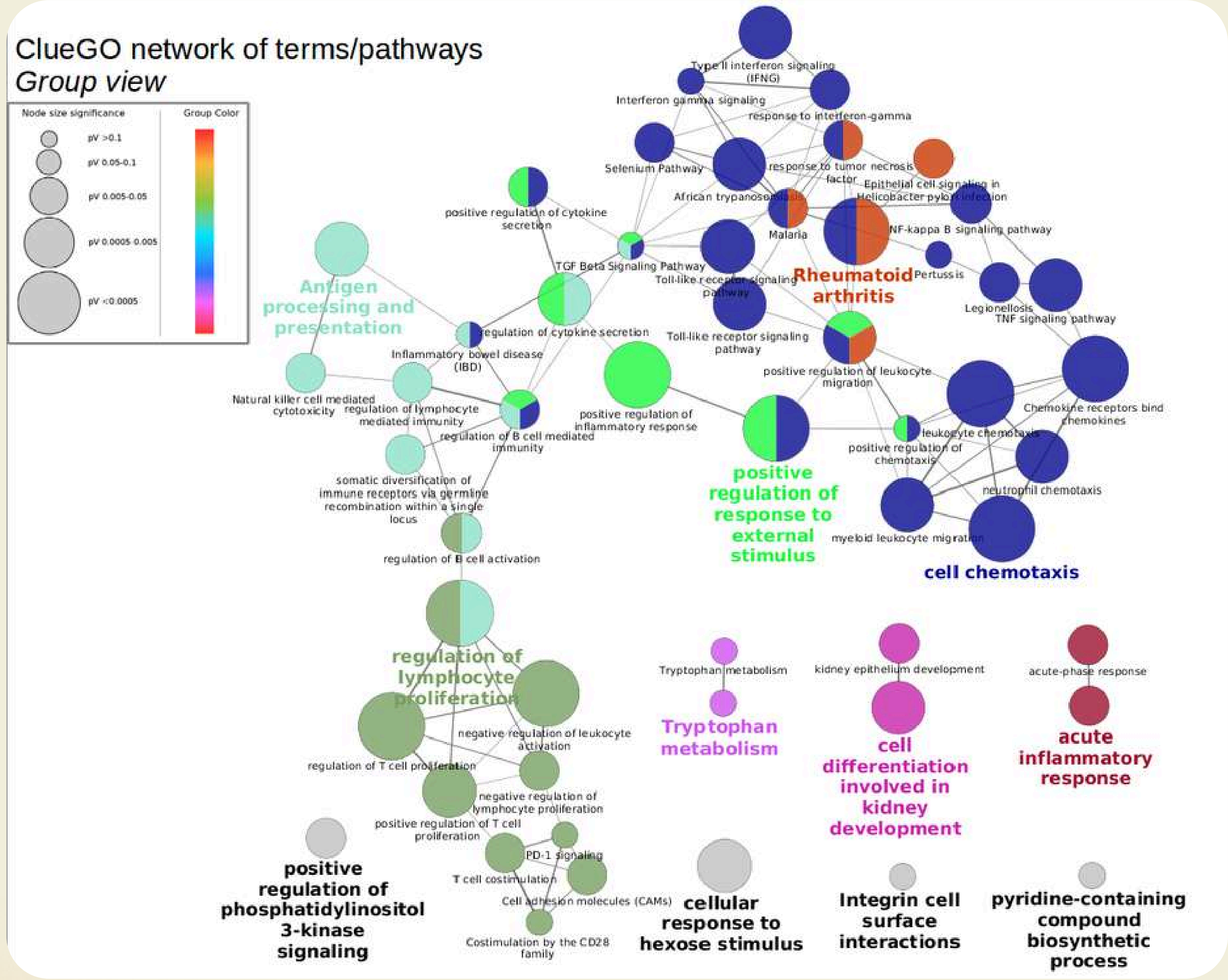
Gene Ontology (GO)

- For each term, it counts how many genes in your list carry that annotation, vs. how many you'd expect if genes were drawn at random from the background
- Statistical test (e.g., hypergeometric/Fisher's exact test): “If I picked this many genes at random from the background, what's the probability I'd see at least this many with this annotation by chance?”
- A small p-value = the overlap is unlikely to be due to chance = the term is “enriched”
- Multiple testing correction (e.g., Bonferroni, Benjamini-Hochberg) is applied because hundreds or thousands of terms are tested at once. Without correction, many “significant” hits would just be false positives



ClueGO: A functional enrichment tool

- A Cytoscape plugin that runs the GO/KEGG/Reactome enrichment test on a gene list.
- Unlike a flat enrichment table (e.g., STRING's built-in tab), ClueGO groups redundant/overlapping terms into functional clusters and draws the result as its own network
- CluePedia is ClueGO's companion app, it adds pathway/disease/miRNA associations and improves term coverage

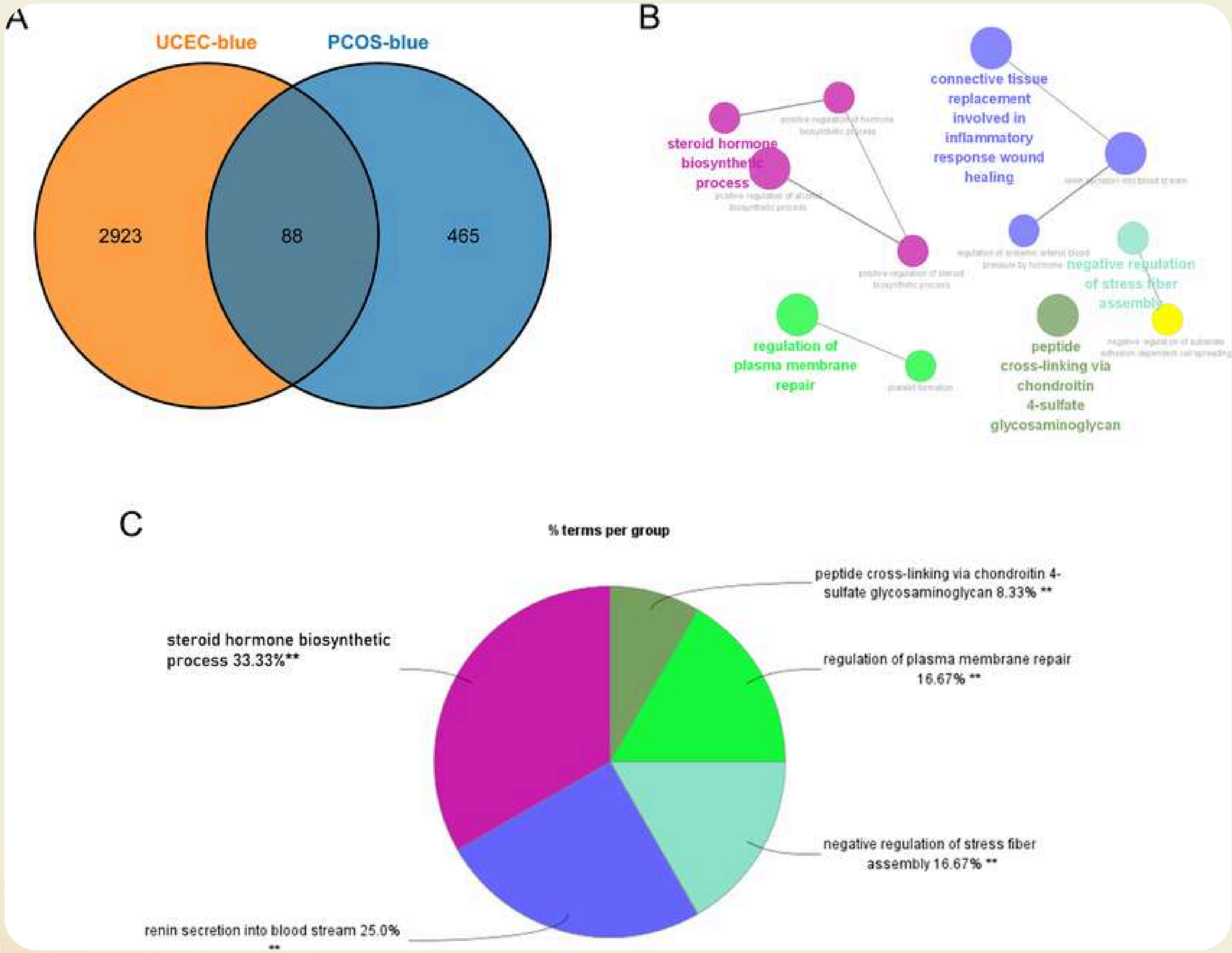


ClueGO: How it works

- Input a gene list
- Choose ontologies/pathways to test against: GO Biological Process, GO Molecular Function, KEGG, Reactome, WikiPathways
- Set a significance threshold (default p-value $\leqslant$ 0.05, corrected)
- ClueGO merges terms that share most of their genes into a single functional group, then colors and labels the network by group, this is what prevents the “200 nearly-identical GO terms” problem of flat enrichment tables

ClueGO: Interpreting the output

- Each node is an enriched term; node size reflects the number of your genes annotated to it
- Term clusters (same color) represent one coherent biological theme, e.g., “cytokine-mediated signaling,” “leukocyte activation”
- A module dominated by one or two clear themes is a strong, coherent result; a scatter of unrelated single-gene terms is a weaker signal
- The Pie Chart view breaks down, per gene, which functional groups it contributes to, this is useful for spotting which specific gene is driving a given theme.



Session task 1

01 In the STRING disease query, search for Crohn's disease and load the network

02 Re-analyze your network using MCODE as previously taught

03 Select the top cluster → create a new network using that cluster → copy the gene names from the node table

04 Open ClueGO → Paste your genes → submit the analysis

STRING enrichment

- Each row is a term (GO/KEGG/Reactome/Pfam), with gene count, background count, and FDR-corrected p-value
- Sort by FDR to see the strongest signals first
- A module dominated by one or two repeating themes across the top rows = coherent; a scatter of unrelated single-hit terms = weaker signal

Session task 2

01 In the STRING disease query, search for Crohn's disease and load the network

02 Go to Apps → STRING enrichment → Retrieve functional enrichment

03 Sort by FDR value, ascending

04 Take a look at the description column, is there a common theme?

WikiPathways

- A community-curated, open pathway database – like Reactome/KEGG, but community-editable and versioned
- The WikiPathways Cytoscape app imports an actual curated pathway diagram as a live, analyzable network – not just an enrichment statistic
- Search by gene, disease, or pathway name directly from within Cytoscape (requires an internet connection to the WikiPathways web service)

What makes a target druggable?

- **Structural tractability:** a solved 3D structure exists, with an identifiable binding pocket.
- Precedent: known ligands or inhibitors already exist against this protein or its close homologs.
- **Disease relevance:** strong, reproducible genetic/expression evidence linking the target to the disease.
- **Network centrality:** consistent hub/bottleneck status across multiple independent metrics.
- **Safety considerations:** broadly-expressed essential genes carry higher off-target risk than tissue-restricted or disease-specific ones.
- Most drug targets are enzymes and receptors

CyTargetLinker

- A Cytoscape app that overlays curated regulatory and drug-target links directly onto your existing network
- Works from pre-built “linksets”:curated edge collections sourced from DrugBank, the Therapeutic Target Database (TTD), and others
- Extends your network with new nodes: known drugs/compounds connected to any of your genes that they target
- It answers the question: “has this candidate ever been drugged before?” visually, inside the same network you've already built without the need for separate database lookup needed.

Session task 3

01 On the CyTargetLinker website, download the DrugBank listset .

02 On the Crohn’s disease network, Apps → CyTargetLinker → Extract Link Sets → Choose the DrugBank linkset (human).

03 Extend the network

04 Change layout to circular. Does your hub protein have an existing drug?

Project task 3

01 Apps → CyTargetLinker → Extract Link Sets → DrugBank (human)

02 Apps → CyTargetLinker → Extend Network → select your disease network and the DrugBank linkset you just extracted → run.

03 From your consensus hub shortlist, choose one protein with zero DrugBank connections. This is your final target candidate.

04 Discuss your protein in 1-2 paragraphs.

Summary

Question	Tool to answer the question
What does cluster 1 do?	ClueGo, STRING enrichment
Has any candidate gene ever been drugged before?	CyTargetLinker

Thank You

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